

Shuvam Banerji Seal

BS-MS Student (Chemistry Major, CS Minor) | IISER Kolkata | Fifth-Year Dual-Degree Candidate
Computational Chemist | AI/ML & Information Retrieval Researcher | DeepTech & AI-Fintech Co-Founder

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Research Focus

Master's Thesis I – Genomic Sequences & Formal Mathematics: Encoding DNA, RNA and protein sequences into binary sequence space on **Gray-code** orderings and **Karnaugh maps**, so map adjacency is a single-symbol mutation; every theorem machine-checked in **Lean 4** (Mathlib).

Master's Thesis II – RNA Structure Prediction: Sparse Mixture-of-Experts **transformer** for RNA secondary and tertiary structure, routing helices, loops and pseudoknots to specialised experts; geometry-aware attention with distance-map and torsion-angle heads.

Computational Chemistry: Density functional theory with **Quantum ESPRESSO** and Gaussian; molecular dynamics in LAMMPS; ML surrogates for simulation.

Core AI/ML & Agentic Systems: Retrieval-augmented generation, **AI safety** and syco-phancy evaluation, agentic AI, information theory.

Publications

AgriIR: Scalable Framework for Domain-Specific Knowledge Retrieval *ECIR 2026*

S. Banerji Seal, A. Poddar, A. Mishra, D. Roy [DOI](#)

SycoLex: Benchmark for Sycophancy in Legal Case Reasoning *CIKM 2026 (under review)*

S. Banerji Seal et al. 1,954 cases, 3 jurisdictions, 5 LLMs. [Dataset](#)

Hierarchical Opinion Classification using LLMs *FIRE 2025*

S. Banerji Seal, A. Mishra, U. Ghosh [Link](#)

Query Reformulation & Layered Retrieval for ToT *TREC 2024*

S. Banerji Seal, S. Adhikary, S. Sar, D. Roy [Link](#)

Computational Modeling of [VO(SALIEP)(DTP)] *To be submitted*

S. Banerji Seal, Dr. S. Roy. DFT (B3LYP) for MO energies.

Flagship Open-Source Software

Molecule3D / LAMMPS Web GUI *2026–*

Browser-native **LAMMPS** workbench: visual script builder over a 186-command library, CMake compiler helper, 3D structure viewer and trajectory analysis (RDF, MSD). TypeScript, React, Three.js. [GitHub](#) · [Live](#)

IISERKonnnect – Campus Super-App *2026–*

Sole developer. Android (Kotlin) + web app unifying the whole IISER-K intranet (WeLearn, mess, calendar, grades, library, notices, network). **In campus-wide testing with 2,000+ users.** [Live](#)

DFT Notes · Intro to Molecular Simulation *2026*

Open knowledge base on density functional theory, and a beginner-to-running-simulation MD course. [Notes](#) · [GitHub](#)

Startups & Freelance

Co-Founder & Lead Developer, iFiNN *2025–Present*

MeitY GENESIS-funded AI-fintech startup. Smart trading alerts for stocks, crypto, mutual funds. [Website](#)

Co-Founder & CTO, UnderWater AI *2025–Present*

MeitY GENESIS-funded DeepTech startup. Deep learning for underwater image enhancement. [Website](#)

Freelance Web Developer – ChemActiva Innovations *2024–25*

Built the public website for a nanocellulose deeptech startup (DST-NIDHI PRAYAS, HDFC Parivartan grantee); also Anicon 3.0 and the EFAML lab page. [chemactiva.com](#)

Hackathons & Awards

UIDAI Data Hackathon 2026 – 1st Prize (Rs. 2L) *Jan–May 2026*

National hackathon by UIDAI/NIC/MeitY. Won among 5,000+ submitting teams (15,000 registered): Aadhaar enrolment and biometric-update patterns by region and demographic. [Link](#) · [Press](#)

Capital One Launchpad – Top 14 / 5,073 Teams *2025*

IndicAgri: multi-modal RAG for Indian agriculture, 20+ languages (Team Fibonacci). [GitHub](#)

StatusCode1 – 1st Rank (GIAN Track) *2024*

AI search engine for GIAN's Abandoned US Patents. Nomic Embeddings + similarity search. [GitHub](#)

StatusCode0 – 1st Rank (MATLAB Track) *2023*

Domestic waste data analysis tool, IIIT-Kalyani.

Smart India Hackathon – Finalist *2024*

Problem IDs: SIH1701 (Context-Aware AI for Courts), SIH1734 (Deep Learning for Air Quality).

National Competitions

- **ChemEnigma** (1st) – 72-hr chemistry contest, IISc Bangalore (2025)
- **All Bengal Chemistry Quiz** (2nd RU), Presidency University (2025)
- **Mimansa** (Zonal Topper) – Math, IISER Pune (2024)
- **NAEST** (Zonal RU) – Experimental Skills, IIT Kanpur (2023)
- **JEE Advanced** (Top 0.1%) | **IAT** (Top 0.06%) | **WBJEE** (Top 0.05%)

Invited Talks & Jury Positions

- **Guest Lecturer & Guest of Honour**, CBSE District-Level Deliberation on Computational Thinking & AI, Kolkata (Aug 2026). Addressed **64 teachers from 38 schools** as an AI subject expert; talk released openly as an [HTML deck](#) and [Beamer PDF](#).
- **Invited Judge**, Voice of Youth 2026 National Summit, Kolkata (1–2 Aug 2026)
- **Track Co-organizer, FIRE 2026 – SYCO PHANCY:** Explainable AI in Legal Reasoning (IISER-K, UBO France, UvA Netherlands)
- **Domain Judge**, La Martiniere for Boys Science Fest, Kolkata (2025)
- **Speaker:** Running & Optimising Local LLMs; Sustainability, AI & Emerging Technologies (IISER-K, 2025)

Leadership & Community

- **President**, Slashdot – Programming & Design Club, IISER-K (2025–). **Elected with a 99% approval vote** of the club membership.
- **Organizer**, Qiskit Fallfest 2025 (IBM Quantum, 1 of 150 globally)
- **Office Bearer**, Valence Chemistry Club, IISER-K
- COVID-19 relief coordinator; Ek-Pehal mentoring initiative

Technical Skills

- **Computational Chemistry:** DFT, **Quantum ESPRESSO**, Gaussian (B3LYP), ORCA, LAMMPS, GROMACS, VMD, OVITO, AIREBO/ReaxFF/EAM force fields, Monte Carlo
- **AI/ML:** PyTorch, TensorFlow, HuggingFace, transformers, mixture-of-experts, LoRA/QLoRA, distributed training, XGBoost, Scikit-learn
- **IR & RAG:** BM25, Lucene, RRF, FAISS, ChromaDB, LangChain, Ollama, vLLM, dense+sparse retrieval
- **Bioinformatics:** RNA/protein structure prediction, PyMol, ChimeraX, AutoDock Vina, sequence analysis
- **Formal Methods:** Lean 4, Mathlib, machine-checked proofs
- **Languages:** Python, C/C++, Kotlin, TypeScript, Java, Rust, Fortran, Lean 4, Bash
- **Web/Dev:** React, Three.js, WebGL, Django, Flask, Docker, Git, CI/CD, Linux, CUDA, Android/Jetpack Compose

Open Datasets & Education

- **Hugging Face** [huggingface.co/ShuvBan](#): **SycoLex** (1,954 legal cases, 3 jurisdictions, CC-BY-4.0) and **AgriIR_dataset** (15,247 Indian agriculture entries, CC-BY-4.0)
- **IISER Kolkata** – BS-MS (Chemistry Major, CS Minor), CGPA 8.2 (2022–2027)
- **Calcutta University** – B.Sc Honours Physics, CGPA 8.308 (2021–22)
- **Reliance Foundation Scholar** (2023) | **Best Young Scientist Speaker** (2019)